

TrES-3b

R-1695

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_210413 · created 2026-08-02T13:35:58+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459317.9173 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.195 +/- 0.013
Transit depth (Rp/Rs)^2	3.79 +/- 0.51 [%]
Orbital Inclination (inc)	82.05 +/- 0.36
Airmass coefficient 1 (a1)	1.069 +/- 0.012
Airmass coefficient 2 (a2)	-0.059 +/- 0.015
Scatter in the residuals of the lightcurve fit is	1.12 %
Best Comparison Star	#2 - [287, 83]
Optimal Aperture	4.12
Optimal Annulus	10.03
Transit Duration (day)	0.0604 +/- 0.0033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.195 ± 0.013 vs 0.16309 (deviation 2.45σ)
Duration fit vs published	0.0604 d vs 0.0567 d (deviation 1.13σ)
Mid-time O–C	-0.4 min
Depth SNR	15.0
Residual scatter	1.12 %

Light curve

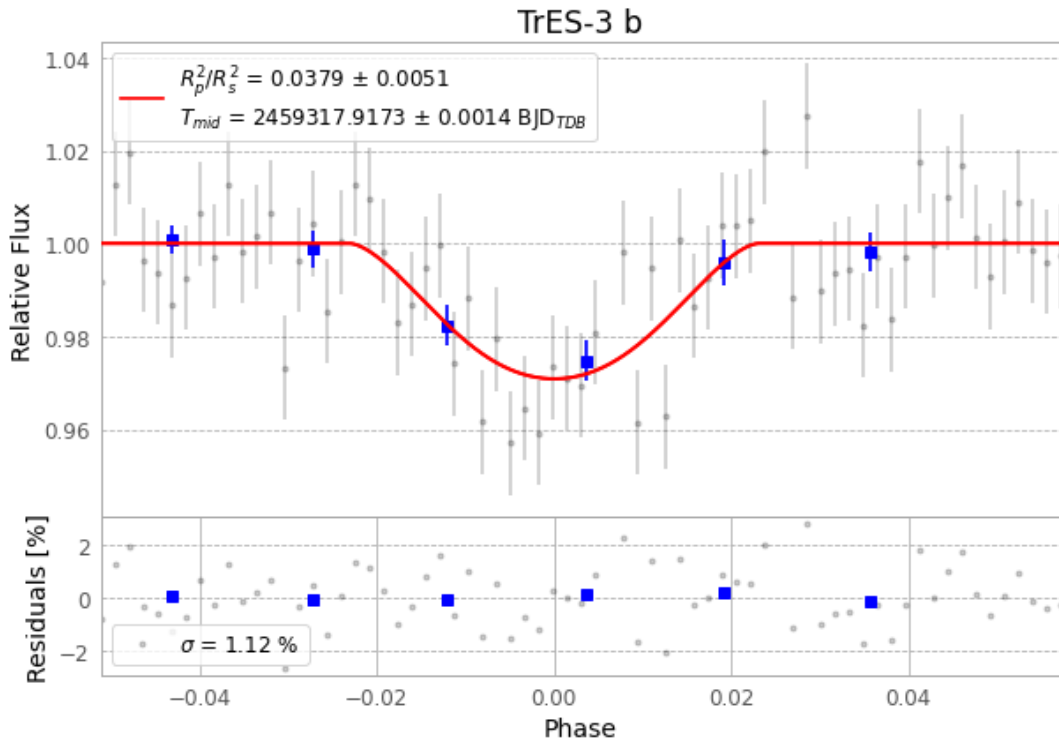


Figure 1 — FinalLightCurve_TrES-3 b_2021-04-13.png (SHA-256 9ae9e8aeb57c8649...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [357, 243], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c5a24c22a0bd22be...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

69 FITS frames (45 MB), TRES-3210413082112.FITS → TRES-3210413114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-210413.log (617cf0e84786...), FinalLightCurve_TrES-3 b_2021-04-13.png (9ae9e8aeb57c...), FinalLightCurve_TrES-3 b_2021-04-13.csv (a46700829aff...)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 13:35Z.